Instructions to the candidates:

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.
4. Assume suitable data if necessary.

Unit III --- Searching and Game Playing

Q1) a) List all problem-solving strategies. What is backtracking? Explain with the N-Queens problem (N=4). [9]

b) Write the Minimax search algorithm for two-player games. Explain how alpha-beta cutoffs improve performance. Apply Minimax with alpha-beta pruning to the following game tree:

```
                 MAX
                /   \
           MIN       MIN
          / | \     / | \
        MAX MAX MAX MAX MAX MAX
        / \ / \ / \ / \ / \ / \
       3  5 6  2 2  1 9  4 7  8
```

[9]

OR

Q2)

a) Define game theory. Differentiate between stochastic games and partial games with examples. [9]

b) What is a Constraint Satisfaction Problem (CSP)? Explain the types of consistencies (node, arc, path). Solve the following cryptarithmetic problem:

```
B A S E

- B A L L

---

G A M E S
```

[8]

Unit IV --- Knowledge Representation

Q3)

a) What is an AI agent? Name any 5 agents from the real world. Explain a knowledge-based agent with reference to the Wumpus World environment. [9]

b) List and explain the various steps of the knowledge engineering process. Consider the following axioms:

> If a triangle is equilateral then it is isosceles.
> If a triangle is isosceles, then its two sides AB and AC are equal.
> If AB and AC are equal, then angle B and angle C are equal.
> ABC is an equilateral triangle.

Represent these facts in First-Order Logic (FOL) and derive that angle B = angle C. [9]

OR

Q4)

a) Write the following sentences in FOL using appropriate quantifiers:

i) All birds fly.

ii) Some boys play cricket.

iii) A first cousin is a child of a parent's sibling.

iv) You can fool all the people some of the time, and some of the people all the time, but you cannot fool all the people all the time. [9]

b) What is knowledge representation using propositional logic? Compare propositional logic and predicate logic (FOL) on at least 8 parameters. [8]

Unit V — Planning and Decision Making

Q5)

a) Explain forward chaining and backward chaining. Discuss their properties, advantages, and disadvantages. [9]

b) Explain:

i) Unification in FOL

ii) Reasoning with default information [9]

OR

Q6)

a) Explain FOL inference rules for the following quantifiers:

i) Universal Generalization

ii) Universal Instantiation

iii) Existential Instantiation

iv) Existential Introduction [9]

b) What is ontological engineering? Explain with detailed categories: objects and model. [9]

Unit VI — AI Applications and Advances

Q7)

a) Explain the Goal Stack Planning algorithm (STRIPS) with a suitable example. Consider the block world problem with initial state: ON(C, A), ON(A, Table), ON(B, Table), CLEAR(B), CLEAR(C). Goal: ON(B, C), ON(C, A). [6]

b) Explain with an example how planning differs from problem solving. [5]

c) Explain AI components and AI architecture. [8]

OR

Q8)

a) Explain planning in non-deterministic domains. [5]

b) Explain:

i) Importance of planning in AI

ii) Algorithm for classical planning [5]

c) What is AI? Explain the scope of AI in various walks of life and discuss future opportunities with AI. [8]

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Examiner Commentary

This paper follows the SPPU AI pattern seen in actual PYQs. Q1(a-b) tests game theory and search with a numerical alpha-beta pruning tree. Q2(b) involves a real cryptarithmetic puzzle. Q3(b) tests FOL representation and logical derivation. Q7(a) involves STRIPS planning with block world --- a classic AI problem type.

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